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# ============================================================================
# CCS2213 – Machine Learning  |  Project #7
# Employee Productivity Analysis and Attrition Prediction
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# STEP 1 – DATA PREPROCESSING
# Deliverables: #2 (EDA), #3 (Missing Values), #4 (Scaling),
#               #5 (Encoding), #6 (Feature Engineering + Split)
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# ── 0. INSTALL & IMPORT ──────────────────────────────────────────────────
# !pip install pandas numpy matplotlib seaborn scikit-learn --quiet

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split

import warnings
warnings.filterwarnings('ignore')

# Plot style
sns.set_theme(style="whitegrid", palette="Set2")
plt.rcParams['figure.figsize'] = (10, 5)

print("=" * 60)
print("STEP 1 – DATA PREPROCESSING")
print("=" * 60)
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STEP 1 – DATA PREPROCESSING
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# ── 1. LOAD DATASET ──────────────────────────────────────────────────────
# Download from Kaggle:
# https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
# Upload to Colab or place in same directory

df = pd.read_csv('WA_Fn-UseC_-HR-Employee-Attrition.csv')

print(f"\n[1] Dataset loaded successfully.")
print(f"    Shape: {df.shape[0]} rows × {df.shape[1]} columns")
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[1] Dataset loaded successfully.
    Shape: 1470 rows × 35 columns
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# ── 2. INITIAL EDA ───────────────────────────────────────────────────────
print("\n" + "─" * 60)
print("[2] INITIAL EXPLORATORY DATA ANALYSIS")
print("─" * 60)

print("\n>> Data Types & Non-Null Counts:")
print(df.info())

print("\n>> First 5 Rows:")
print(df.head())

print("\n>> Descriptive Statistics (Numerical Columns):")
print(df.describe().round(2))

print("\n>> Target Variable Distribution:")
attrition_counts = df['Attrition'].value_counts()
attrition_pct    = df['Attrition'].value_counts(normalize=True) * 100
print(attrition_counts)
print(f"\n    No  (0): {attrition_pct['No']:.1f}%")
print(f"    Yes (1): {attrition_pct['Yes']:.1f}%")
print("\n    NOTE: Dataset is imbalanced (~84% No, ~16% Yes).")
print("          class_weight='balanced' will be used in all models.")

# Plot target distribution
fig, ax = plt.subplots(figsize=(6, 4))
attrition_counts.plot(kind='bar', color=['#2E7D32', '#C62828'], edgecolor='white', ax=ax)
ax.set_title('Target Variable Distribution – Attrition', fontsize=13, fontweight='bold')
ax.set_xlabel('Attrition')
ax.set_ylabel('Count')
ax.set_xticklabels(['No', 'Yes'], rotation=0)
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for p in ax.patches:
    ax.annotate(f'{int(p.get_height())}', (p.get_x() + p.get_width() / 2, p.get_height()),
                ha='center', va='bottom', fontsize=11)
plt.tight_layout()
plt.savefig('fig_target_distribution.png', dpi=150)
plt.show()
print("    [Saved] fig_target_distribution.png")

# Categorical columns overview
cat_cols = df.select_dtypes(include='object').columns.tolist()
num_cols = df.select_dtypes(include=['int64', 'float64']).columns.tolist()
print(f"\n>> Categorical columns ({len(cat_cols)}): {cat_cols}")
print(f">> Numerical columns  ({len(num_cols)}): {num_cols}")
```

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[2] INITIAL EXPLORATORY DATA ANALYSIS

>> Data Types & Non-Null Counts:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1470 non-null   int64
 1   Attrition                 1470 non-null   object
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64
 6   Education                 1470 non-null   int64
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64
 9   EmployeeNumber            1470 non-null   int64
 10  EnvironmentSatisfaction   1470 non-null   int64
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64
 13  JobInvolvement            1470 non-null   int64
 14  JobLevel                  1470 non-null   int64
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64
 19  MonthlyRate               1470 non-null   int64
 20  NumCompaniesWorked        1470 non-null   int64
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64
 24  PerformanceRating         1470 non-null   int64
 25  RelationshipSatisfaction  1470 non-null   int64
 26  StandardHours             1470 non-null   int64
 27  StockOptionLevel          1470 non-null   int64
 28  TotalWorkingYears         1470 non-null   int64
 29  TrainingTimesLastYear     1470 non-null   int64
 30  WorkLifeBalance           1470 non-null   int64
 31  YearsAtCompany            1470 non-null   int64
 32  YearsInCurrentRole        1470 non-null   int64
 33  YearsSinceLastPromotion   1470 non-null   int64
 34  YearsWithCurrManager      1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
None

>> First 5 Rows:
   Age Attrition     BusinessTravel  DailyRate              Department  \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
3   33        No  Travel_Frequently       1392  Research & Development
4   27        No      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber  \
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7

   ...  RelationshipSatisfaction StandardHours  StockOptionLevel  \
0  ...                         1            80                 0
1  ...                         4            80                 1
2  ...                         2            80                 0
3  ...                         3            80                 0
4  ...                         4            80                 1

   TotalWorkingYears  TrainingTimesLastYear WorkLifeBalance  YearsAtCompany  \
0                  8                      0               1               6
1                 10                      3               3              10
2                  7                      3               3               0
3                  8                      3               3               8
4                  6                      3               3               2

  YearsInCurrentRole  YearsSinceLastPromotion  YearsWithCurrManager
0                  4                        0                     5
1                  7                        1                     7
2                  0                        0                     0
3                  7                        3                     0
4                  2                        2                     2

[5 rows x 35 columns]

>> Descriptive Statistics (Numerical Columns):
           Age  DailyRate  DistanceFromHome  Education  EmployeeCount  \
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       count  1470.00    1470.00           1470.00    1470.00         1470.0
       mean     36.92     802.49              9.19       2.91            1.0
       std       9.14     403.51              8.11       1.02            0.0
       min      18.00     102.00              1.00       1.00            1.0
       25%      30.00     465.00              2.00       2.00            1.0
       50%      36.00     802.00              7.00       3.00            1.0
       75%      43.00    1157.00             14.00       4.00            1.0
       max      60.00    1499.00             29.00       5.00            1.0

              EmployeeNumber  EnvironmentSatisfaction  HourlyRate  JobInvolvement  \
       count         1470.00                  1470.00     1470.00         1470.00
       mean          1024.87                     2.72       65.89            2.73
       std            602.02                     1.09       20.33            0.71
       min              1.00                     1.00       30.00            1.00
       25%            491.25                     2.00       48.00            2.00
       50%           1020.50                     3.00       66.00            3.00
       75%           1555.75                     4.00       83.75            3.00
       max           2068.00                     4.00      100.00            4.00

              JobLevel  ...  RelationshipSatisfaction  StandardHours  \
       count   1470.00  ...                   1470.00         1470.0
       mean       2.06  ...                      2.71           80.0
       std        1.11  ...                      1.08            0.0
       min        1.00  ...                      1.00           80.0
       25%        1.00  ...                      2.00           80.0
       50%        2.00  ...                      3.00           80.0
       75%        3.00  ...                      4.00           80.0
       max        5.00  ...                      4.00           80.0

              StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  \
       count           1470.00            1470.00                1470.00
       mean               0.79              11.28                   2.80
       std                0.85               7.78                   1.29
       min                0.00               0.00                   0.00
       25%                0.00               6.00                   2.00
       50%                1.00              10.00                   3.00
       75%                1.00              15.00                   3.00
       max                3.00              40.00                   6.00

              WorkLifeBalance  YearsAtCompany  YearsInCurrentRole  \
       count          1470.00         1470.00             1470.00
       mean              2.76            7.01                4.23
       std               0.71            6.13                3.62
       min               1.00            0.00                0.00
       25%               2.00            3.00                2.00
       50%               3.00            5.00                3.00
       75%               3.00            9.00                7.00
       max               4.00           40.00               18.00

              YearsSinceLastPromotion  YearsWithCurrManager
       count                  1470.00               1470.00
       mean                      2.19                  4.12
       std                       3.22                  3.57
       min                       0.00                  0.00
       25%                       0.00                  2.00
       50%                       1.00                  3.00
       75%                       3.00                  7.00
       max                      15.00                 17.00

       [8 rows x 26 columns]

       >> Target Variable Distribution:
       Attrition
       No     1233
       Yes     237
       Name: count, dtype: int64

           No  (0): 83.9%
           Yes (1): 16.1%

           NOTE: Dataset is imbalanced (~84% No, ~16% Yes).
                 class_weight='balanced' will be used in all models.
```

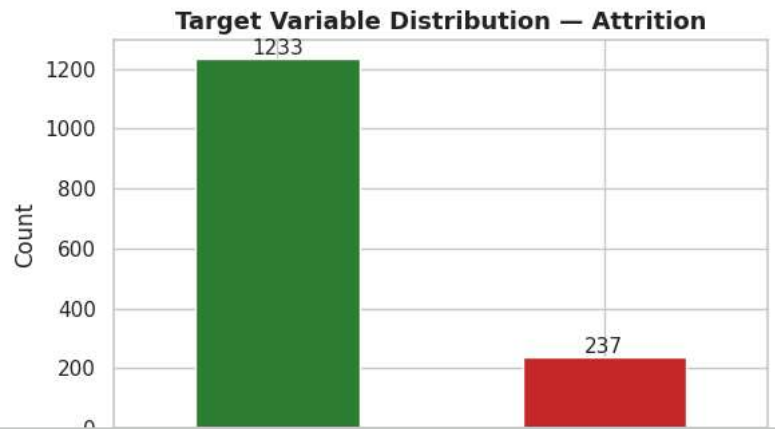

```
# ── 3. MISSING VALUE ANALYSIS ──────────────────────────────────────────
print("\n" + "─" * 60)
print("[3] MISSING VALUE ANALYSIS")
print("─" * 60)

missing = df.isnull().sum()
missing_pct = (missing / len(df)) * 100
missing_df = pd.DataFrame({'Missing Count': missing, 'Missing %': missing_pct})
missing_df = missing_df[missing_df['Missing Count'] > 0]

if missing_df.empty:
    print("\n    No missing values found. Dataset is clean.")
    print("    No imputation or removal required.")
else:
    print("\n    Columns with missing values:")
    print(missing_df)
```

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[3] MISSING VALUE ANALYSIS
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    No missing values found. Dataset is clean.
    No imputation or removal required.
```

```
# ── 4. FEATURE ENGINEERING ─────────────────────────────────────────────
print("\n" + "─" * 60)
print("[4] FEATURE ENGINEERING")
print("─" * 60)

# Drop constant / non-informative columns
cols_to_drop = ['EmployeeCount', 'StandardHours', 'Over18', 'EmployeeNumber']
df.drop(columns=cols_to_drop, inplace=True)
print(f"\n    Dropped non-informative columns: {cols_to_drop}")
print(f"    Reason: These columns have constant values and provide no")
print(f"            predictive signal for the model.")
print(f"\n    New shape: {df.shape[0]} rows × {df.shape[1]} columns")
```

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[4] FEATURE ENGINEERING
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    Dropped non-informative columns: ['EmployeeCount', 'StandardHours', 'Over18', 'EmployeeNumber']
    Reason: These columns have constant values and provide no
            predictive signal for the model.

    New shape: 1470 rows × 31 columns
```

```
# ── 5. ENCODE CATEGORICAL VARIABLES ────────────────────────────────────
print("\n" + "─" * 60)
print("[5] ENCODING CATEGORICAL VARIABLES")
print("─" * 60)

# 5a. Label encode binary categoricals
le = LabelEncoder()
binary_cols = ['Attrition', 'OverTime', 'Gender']
for col in binary_cols:
    df[col] = le.fit_transform(df[col])
    print(f"    LabelEncoder → {col}  (classes: {list(le.classes_)} → {list(range(len(le.classes_)))})")

print(f"\n    Reason: Binary columns only have 2 values — LabelEncoder")
print(f"            converts them to 0/1 without creating unnecessary columns.")

# 5b. One-Hot Encode multi-class categoricals
ohe_cols = ['Department', 'JobRole', 'MaritalStatus', 'EducationField', 'BusinessTravel']
df = pd.get_dummies(df, columns=ohe_cols, drop_first=True)
print(f"\n    One-Hot Encoding applied to: {ohe_cols}")
print(f"    Reason: Multi-class nominal columns — no ordinal relationship")
print(f"            exists, so integer encoding would imply false ordering.")
print(f"\n    Shape after encoding: {df.shape[0]} rows × {df.shape[1]} columns")
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────────────────────────────────────────────────────────────
[5] ENCODING CATEGORICAL VARIABLES
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    LabelEncoder → Attrition  (classes: ['No', 'Yes'] → [0, 1])
    LabelEncoder → OverTime  (classes: ['No', 'Yes'] → [0, 1])
    LabelEncoder → Gender  (classes: ['Female', 'Male'] → [0, 1])
```